

Operator and observer identity checks

Signal Space / GROSS Test 1

Classification: pass. Technical execution: completed.

Question: are the operator and observer identities implemented consistently, including nonunitary frame changes?

Measured: 1000 seeded samples; maximum scale-normalized residual $3.96312\text{e-}14$; locked threshold $1\text{e-}10$.

Tetrahedral spectrum error: $4.44089\text{e-}16$. Omitted-weight negative-control error: 0.632121.

Method: saved complex 2×2 matrices and state vectors are audited using independent real Pauli components. Random Hermitian matrices and positive aggregates have condition number at most 100. $SL(2, \mathbb{C})$ transformations combine $SU(2)$ rotations with boosts of rapidity between -1 and 1.

The exact rank-one aggregate has no normalized timelike observer. Near-collinear controls are reported separately, without clamping eigenvalues or inventing finite observer values.

Error budget: floating-point algebra and conditioning; no fitted parameters, smoothing or stochastic confidence claim. Spatial boundary, conservation and timestep convergence are not applicable to this non-evolving matrix experiment.

Limits: validates algebra/implementation only. No physical time, propagation, bound clock, Lorentz covariance of dynamics, or emergent spacetime is established.

Next: Test 2 reciprocal event conservation and permissible rescheduling; Test 5 continuum action audit is an independent branch.

Run: run-2ae9c65dbd171851

Analysis: analysis-0001-d3c876ea

Solver commit: adbe7584dbd8b13b34432cbd88b5a6c8f7e39709

Config SHA-256: b40ef9c9fa4f0f03640acfe953270e3241c425f72438bcb5983112c33f01f21b

Renderer commit: b99885fdc5967ebe7625d5efcbd5aa6c04830b92

operator-residuals

Question: Do operator and observer identities remain accurate across the declared conditioning domain?

Reading: Maximum residual $3.96\text{e-}14$; compare all points with the locked $1\text{e-}10$ line. Log plotting floors exact zeros at $1\text{e-}18$.

Significance: Tests consistent determinant, observer norm, positive form and weighted-ray transformations on saved inputs, with an independent component audit.

Limitation: Finite moderate-conditioning samples; no propagation, clock or emergent metric is measured.

tetrahedral-gram

Question: Do four null projector directions span the predicted Lorentzian operator space?

Reading: Each projector has a null diagonal. The sorted spectrum matches $(-1/3, -1/3, -1/3, 1)$ with maximum error $4.44\text{e-}16$.

Operator and observer identity checks

Significance: Four null projector directions can span the four-dimensional Hermitian operator space with Lorentzian signature.

Limitation: This is internal algebra. The tetrahedron supplies neither physical spatial axes nor a dynamical spacetime.

observer-domain

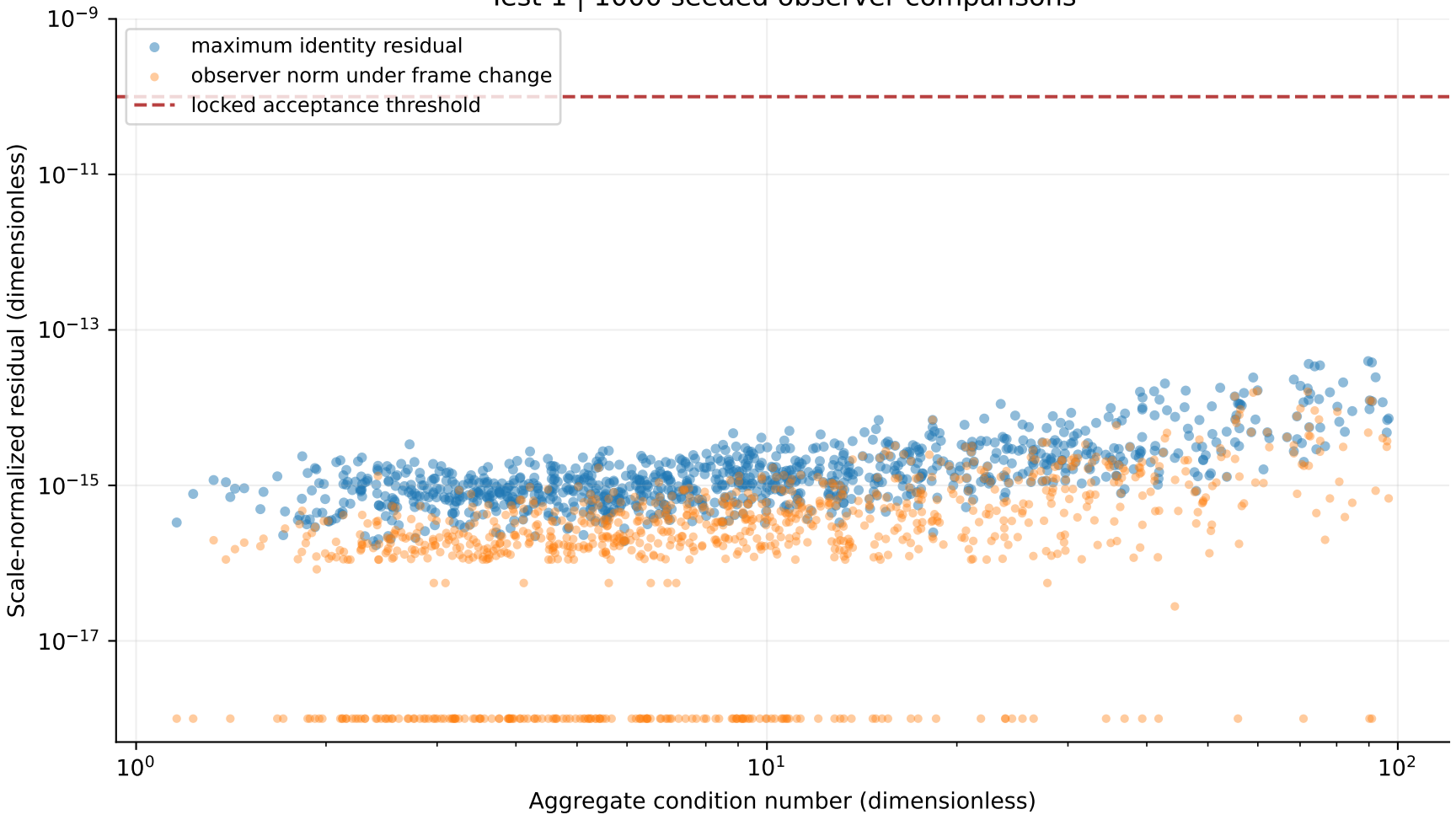
Question: Do singular aggregates and omitted ray weights expose the intended failure controls?

Reading: Collinearity is rejected and ill-conditioned aggregates are marked separately. Omitting a boost-transformed weight gives error 0.632121.

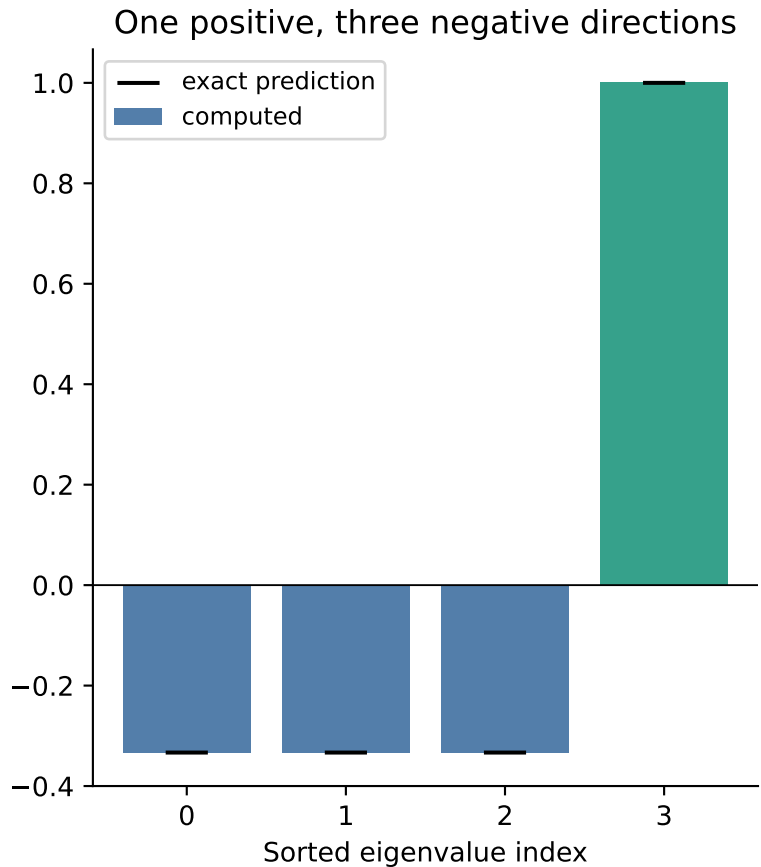
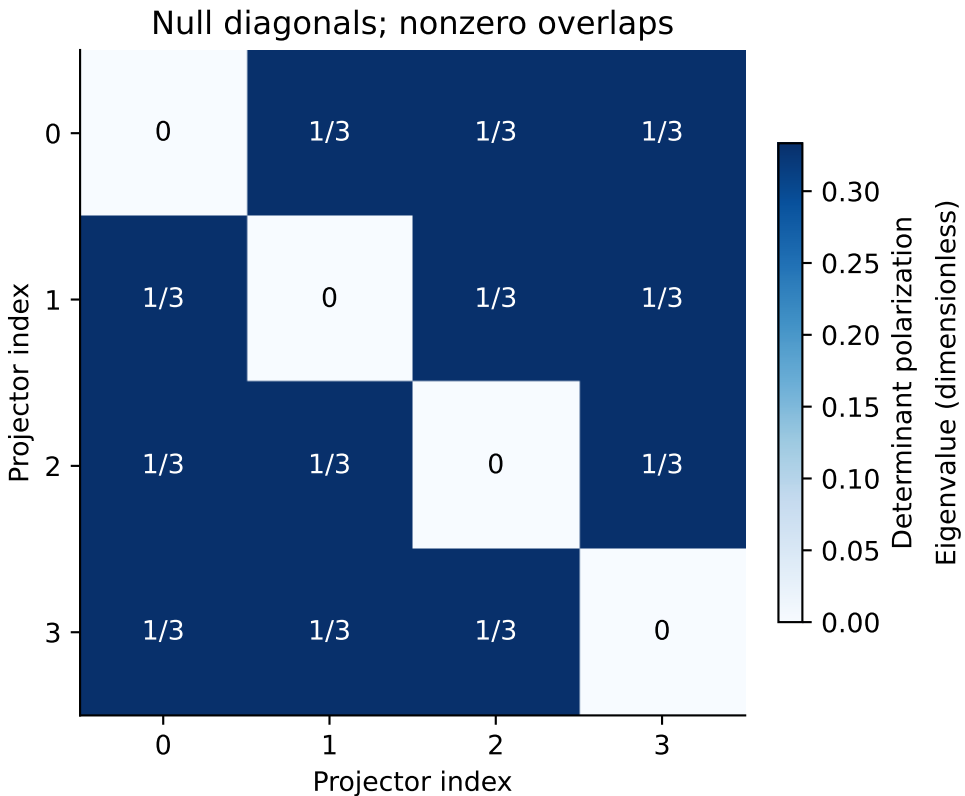
Significance: The test distinguishes valid observer construction from a singular ray, and detects inconsistent transformation of physical weights.

Limitation: A domain rejection is not a physical instability. The deliberately broken control is not an alternative dynamics.

Test 1 | 1000 seeded observer comparisons

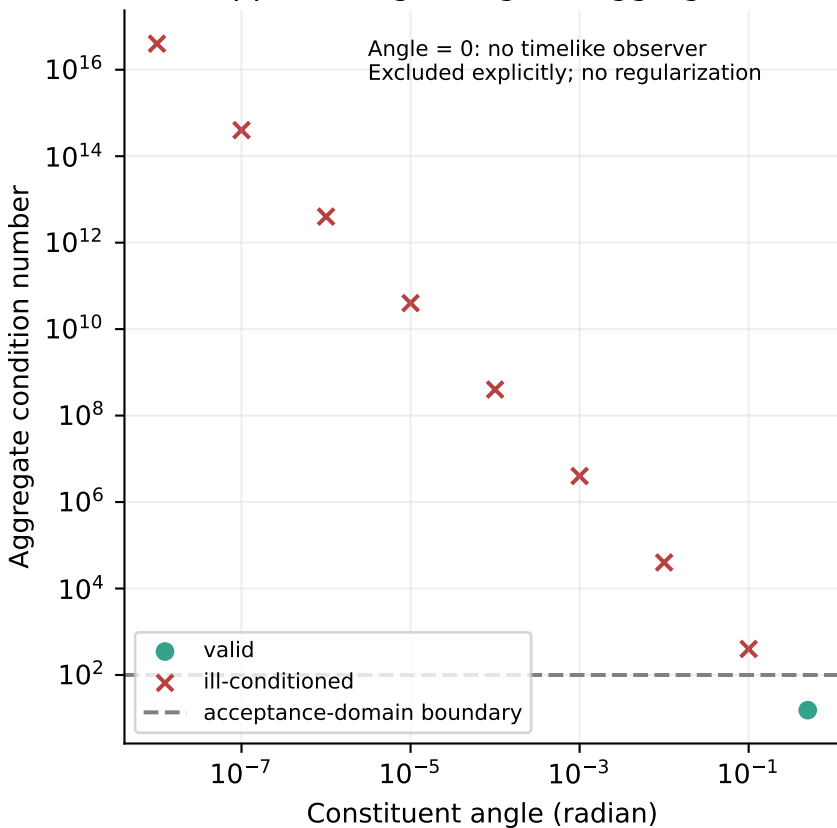


Test 1 | Algebraic signature, not a propagation measurement



Test 1 | Invalid domains and deliberately inconsistent transformations

Approaching a singular aggregate



Same nonunitary frame change

